

# Day 45 - NAT (pt 2)

Wednesday, September 9, 2026

8:37 PM

## NAT Info

### Static NAT Reminder

#### Static NAT

**Static** NAT involves **statically** configuring **one-to-one** mappings of **private** IP addresses to **public** IP addresses.

- When traffic from the internal host is sent to the outside network, the router will translate the **source** address.
- Also, this **one-to-one** mapping also allows external hosts to access the internal host via the **Inside Global** address.

#### Static NAT Configuration

- **show ip nat translations**
  - Shows information about IP addresses that have been translated through **NAT**.
  - Includes the following information:
    - **Protocol** used in the connection
    - **Inside Local** address
    - **Inside Global** address
    - **Outside Local** address
    - **Outside Global** address
- **show ip nat statistics**
  - Shows statistics about the current **NAT** process
  - Includes the following information:
    - Total active translations (**static**, **dynamic**, **extended**)
    - Peak translations (highest amount of translations in one time)
    - **Outside** interfaces
    - **Inside** interfaces
    - Etc.
- **clear ip nat translation \***
  - Clears all **dynamic** **NAT** translations in the **NAT** translation table
  - (One type of **dynamic** translations is IP address translations with a port included)
- **ip nat inside**
  - Defines the '**inside**' interface(s) connected to the **internal** network
- **ip nat outside**
  - Defines the '**outside**' interface(s) connected to the **external** network
- **ip nat source static INSIDE-LOCAL-IP INSIDE-GLOBAL-IP**
  - Configures a one-to-one (**Static NAT**) IP address mapping

[Link to original](#)

### Dynamic NAT

**Dynamic** NAT involves **dynamically** mapping **Inside Local** addresses to **Inside Global** addresses as needed.

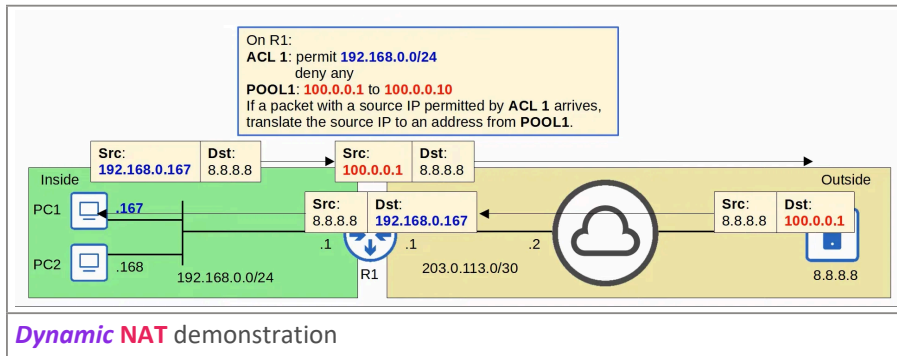
- Although they are **dynamically** assigned, the mappings are still **one-to-one**. (One **Inside Local** IP address per **Inside Global** IP address)
- It uses an **ACL** to identify which traffic should be **translated**.
  - If the source IP is **permitted** by the **ACL**, the source IP will be **translated**.
  - If the source IP is **denied** by the **ACL**, the source IP will not be **translated**.
- A **NAT Pool** is also used to define the available **Inside Global** addresses to be used for the **translation**.

What does it mean if an address is 'not translated'?

- When an address is not **translated** (due to [NAT Pool Exhaustion](#) or being denied by the **ACL**), that does NOT mean that the traffic will not be denied.

mean that the traffic will get dropped.

- Instead, it simply means that the traffic won't go through the **NAT** process and will be sent with its original *source* IP address.



## NAT Pool Exhaustion

If there aren't enough *Inside Global* IP addresses available in the NAT *Pool* (all the addresses are currently being used), that is called '*NAT Pool Exhaustion*'

- If a packet from another *inside* host arrives and needs **NAT** but there are no available addresses, the router will *drop* the packet.
- The host will be unable to access outside networks (which requires **NAT translations**) until one of the *Inside Global* IP addresses becomes available.
- **Dynamic NAT** entries will time out automatically if they're not used, or they can be cleared manually through `clear ip nat translation *`

Be careful: 'Not translated' traffic and 'dropped' traffic

- Watch out for these two similar terms.
  - When traffic is '*not translated*' it means that it won't go through the **NAT** process and will retain its original *source* IP address, therefore, and it can still be sent to other devices in the LAN for example.
  - When traffic needs a **NAT translation** (in order to reach outside networks) and the **Dynamic NAT Pool** is full and completely in use, the traffic will be completely '*dropped*' and discarded.

## Dynamic NAT Configuration

- `show ip nat translations`
  - Shows information about IP addresses that have been translated through **NAT**.
  - Includes the following information:
    - *Protocol* used in the connection
    - *Inside Local* address
    - *Inside Global* address
    - *Outside Local* address
    - *Outside Global* address
- `show ip nat statistics`
  - Shows statistics about the current **NAT** process.
  - Includes the following information:
    - Total active translations (*static, dynamic, extended*)
    - Peak translations (highest amount of translations in one time)
    - *Outside* interfaces
    - *Inside* interfaces
    - Etc.
- `clear ip nat translation *`
  - Clears all **dynamic NAT** translations in the **NAT** translation table.
  - (One type of *dynamic* translations is IP address translations with a port included)
- `ip nat inside`
  - Defines the '*inside*' interface(s) connected to the *internal* network.
- `ip nat outside`
  - Defines the '*outside*' interface(s) connected to the *external* network.

- `access-list ACL-ID permit ARGUMENTS`
  - Defines the traffic that should be *translated*
  - (*Permitted* traffic by the **ACL** will be *translated*)
- `ip nat pool POOL-NAME START-IP END-IP { prefix-length PREFIX-LENGTH | netmask SUBNET-MASK }`
  - Defines the **NAT Pool** of *Inside Global* IP addresses.
- `ip nat inside source list ACL-ID pool POOL-NAME`
  - Configures **Dynamic NAT** on the router by mapping the **ACL** to the **NAT Pool**.

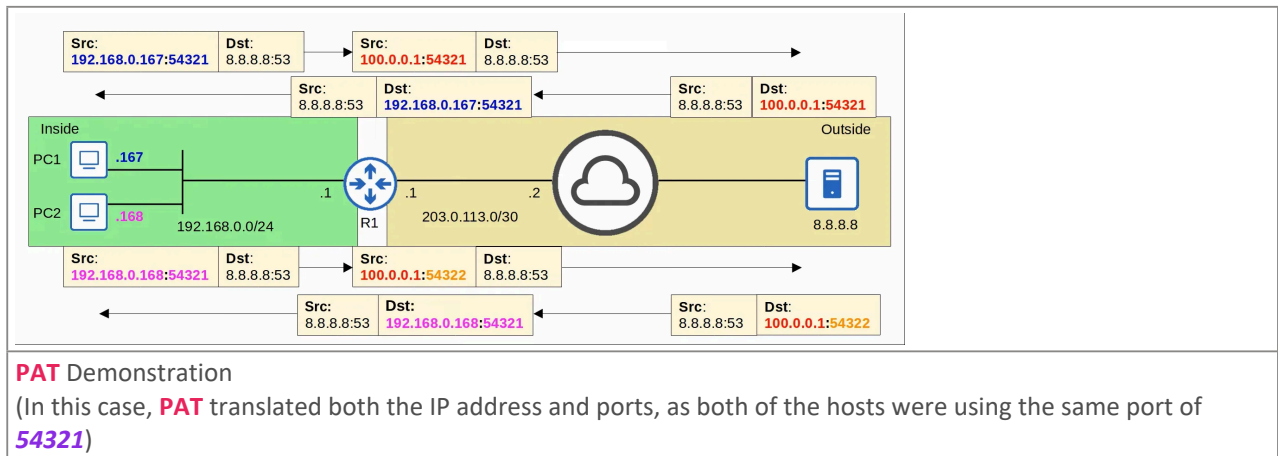
Info: `show ip nat translations` command with **Dynamic NAT**

- When the command is used, the *one-to-one* dynamic mappings will show, just like **Static NAT**, as well as any *translated* addresses and their ports.
- Using the `clear ip nat translation *` command will clear all the *translations* except the *one-to-one* mappings.

## Port Address Translation (PAT) / NAT Overload

Port Address Translation (PAT) or NAT Overload is a protocol that *translates* both the IP address and port number (if necessary)

- By using a unique port number for each communication **flow**, a single public IP address can be used by many different internal hosts.
- (Port numbers are **16 bits**, which is over **65,000** available port numbers)
- The router will keep track of which *Inside Local* address is using which *Inside Global* address and port.
- **PAT** is really good for preserving *public* IP addresses, as many *inside* hosts can share a single *public* IP.



## PAT Configuration (Pools)

- `show ip nat translations`
  - Shows information about IP addresses that have been translated through **NAT**.
  - Includes the following information:
    - *Protocol* used in the connection
    - *Inside Local* address
    - *Inside Global* address
    - *Outside Local* address
    - *Outside Global* address
- `show ip nat statistics`
  - Shows statistics about the current **NAT** process.
  - Includes the following information:
    - Total active translations (*static*, *dynamic*, *extended*)
    - Peak translations (highest amount of translations in one time)
    - *Outside* interfaces
    - *Inside* interfaces
    - Etc.
- `clear ip nat translation *`
  - Clears all *dynamic NAT* translations in the **NAT** translation table.
  - (One type of *dynamic* translations is IP address translations with a port included)

- (One type of **dynamic** translations is IP address translations with a port included)
  - **ip nat inside**
    - Defines the '**inside**' interface(s) connected to the **internal** network.
  - **ip nat outside**
    - Defines the '**outside**' interface(s) connected to the **external** network.
  - **access-list ACL-ID permit ARGUMENTS**
    - Defines the traffic that should be **translated**
    - (**Permitted** traffic by the **ACL** will be **translated**)
  - **ip nat pool POOL-NAME START-IP END-IP { prefix-length PREFIX-LENGTH | netmask SUBNET-MASK }**
    - Defines the **NAT Pool** of **Inside Global** IP addresses.
  - **ip nat inside source list ACL-ID pool POOL-NAME overload**
    - Configures **PAT** on the router by mapping the **ACL** to the **NAT Pool** with the **overload** argument at the end.
- Info:** **show ip nat translations** command with PAT
- When the command is used, only **translated** addresses and their ports will show up.
  - **PAT** does not have any '**one-to-one**' mappings like **Static** and **Dynamic NAT**.
  - Using the **clear ip nat translation \*** command will clear all the **translations**.

## PAT Configuration (Interface)

- **show ip nat translations**
  - Shows information about IP addresses that have been translated through **NAT**.
  - Includes the following information:
    - **Protocol** used in the connection
    - **Inside Local** address
    - **Inside Global** address
    - **Outside Local** address
    - **Outside Global** address
- **show ip nat statistics**
  - Shows statistics about the current **NAT** process.
  - Includes the following information:
    - Total active translations (**static**, **dynamic**, **extended**)
    - Peak translations (highest amount of translations in one time)
    - **Outside** interfaces
    - **Inside** interfaces
    - Etc.
- **clear ip nat translation \***
  - Clears all **dynamic NAT** translations in the **NAT** translation table.
  - (One type of **dynamic** translations is IP address translations with a port included)
- **ip nat inside**
  - Defines the '**inside**' interface(s) connected to the **internal** network.
- **ip nat outside**
  - Defines the '**outside**' interface(s) connected to the **external** network.
- **access-list ACL-ID permit ARGUMENTS**
  - Defines the traffic that should be **translated**
  - (**Permitted** traffic by the **ACL** will be **translated**)
- **ip nat inside source list ACL-ID interface INTERFACE-ID overload**
  - Configures **PAT** on the router by mapping the **ACL** to an interface on the router with the **overload** argument at the end.
  - When traffic is **translated**, it will use the IP address of the specified interface.

